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(54) **SCHEDULING FOR MULTIPLE PRIMARY NODES**

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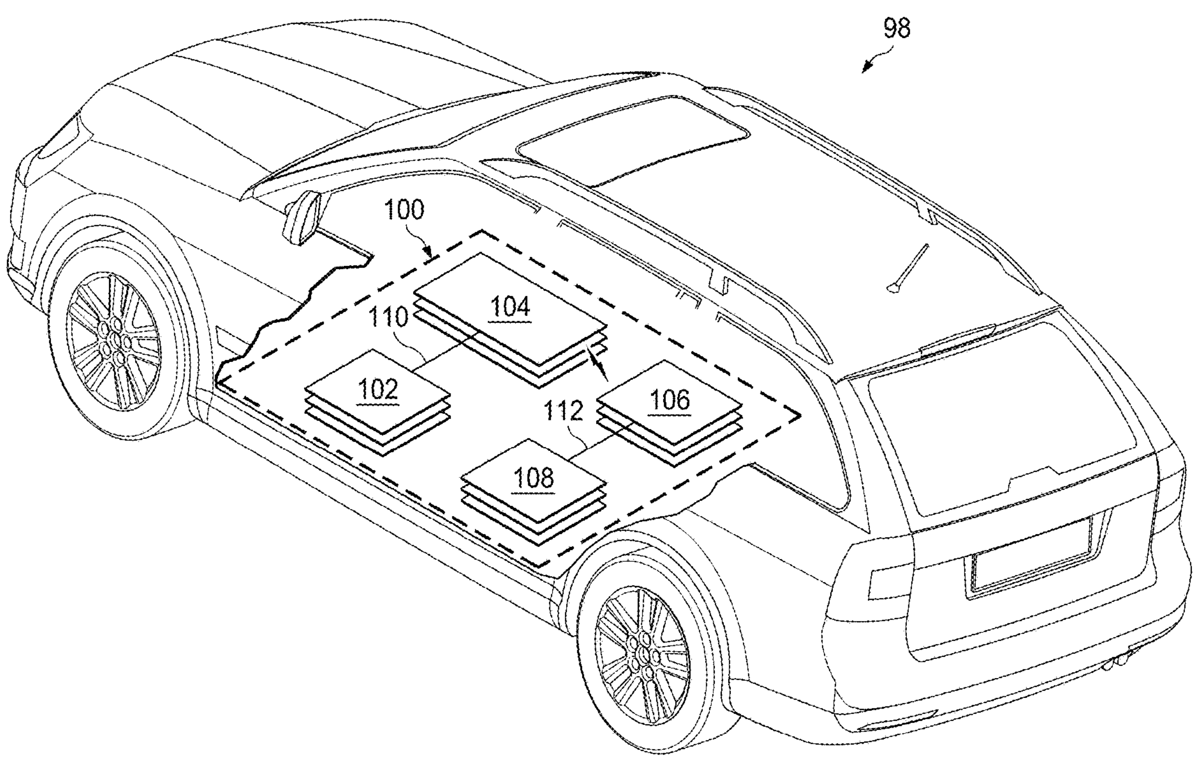
CPC ***H02J 50/80*** (2016.02); ***H04W 72/0446*** (2013.01)

(57) **ABSTRACT**

A device comprises a transceiver that can wirelessly transmit and receive data, and a processor coupled to the transceiver. The processor can receive, from the transceiver and in a first time slot, first data transmitted by a first primary device; receive, from the transceiver and in a second time slot following the first time slot, second data transmitted by a second primary device; provide, to the transceiver and in a third time slot following the second time slot, third data to be transmitted on a first frequency to the first primary device, the third data including a battery cell status; and provide, to the transceiver in a fourth time slot following the second time slot, the third data to be transmitted to the second primary device, the transceiver configured to transmit the third data to the second primary device on a second frequency that is different than the first frequency.



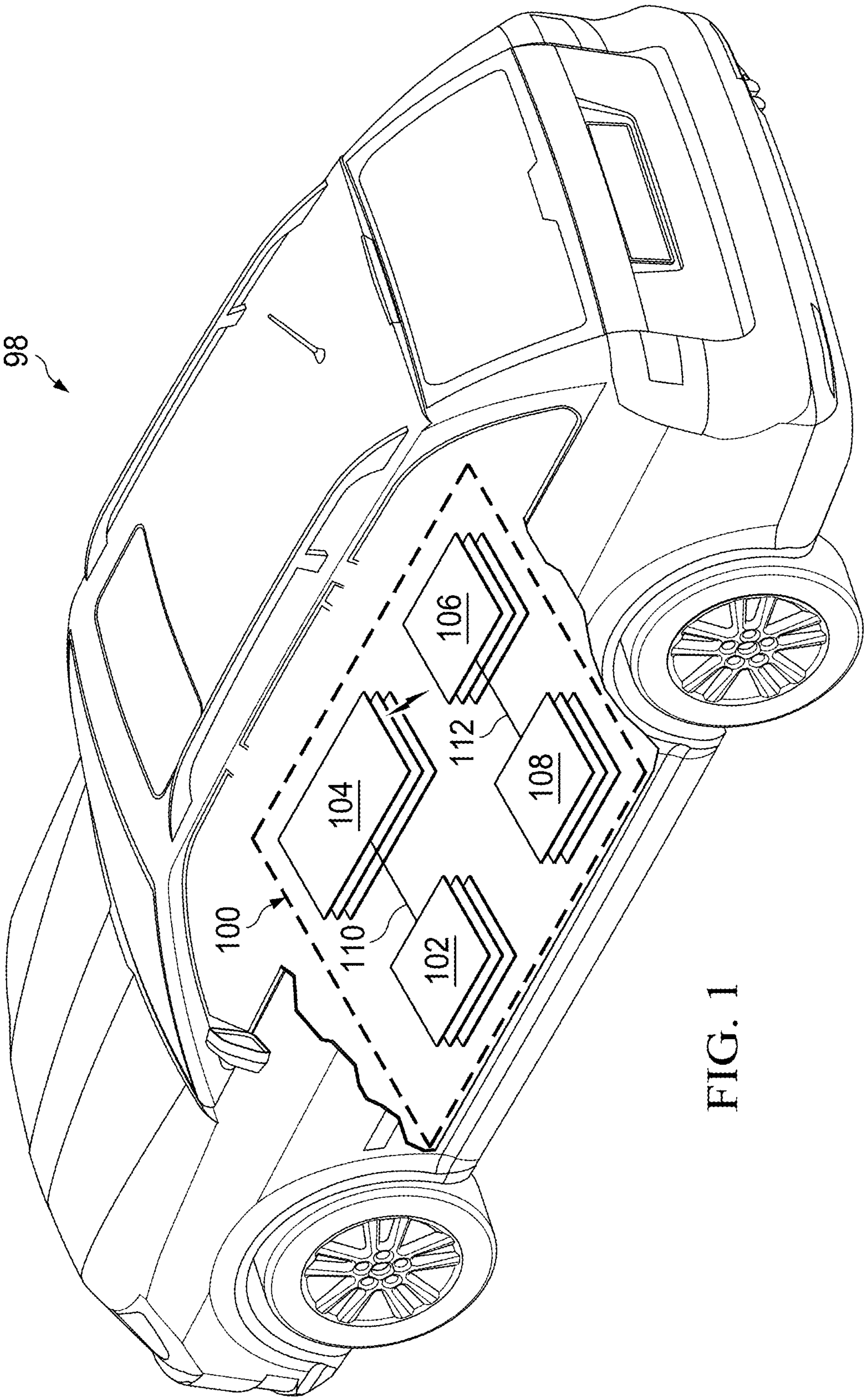


FIG. 1

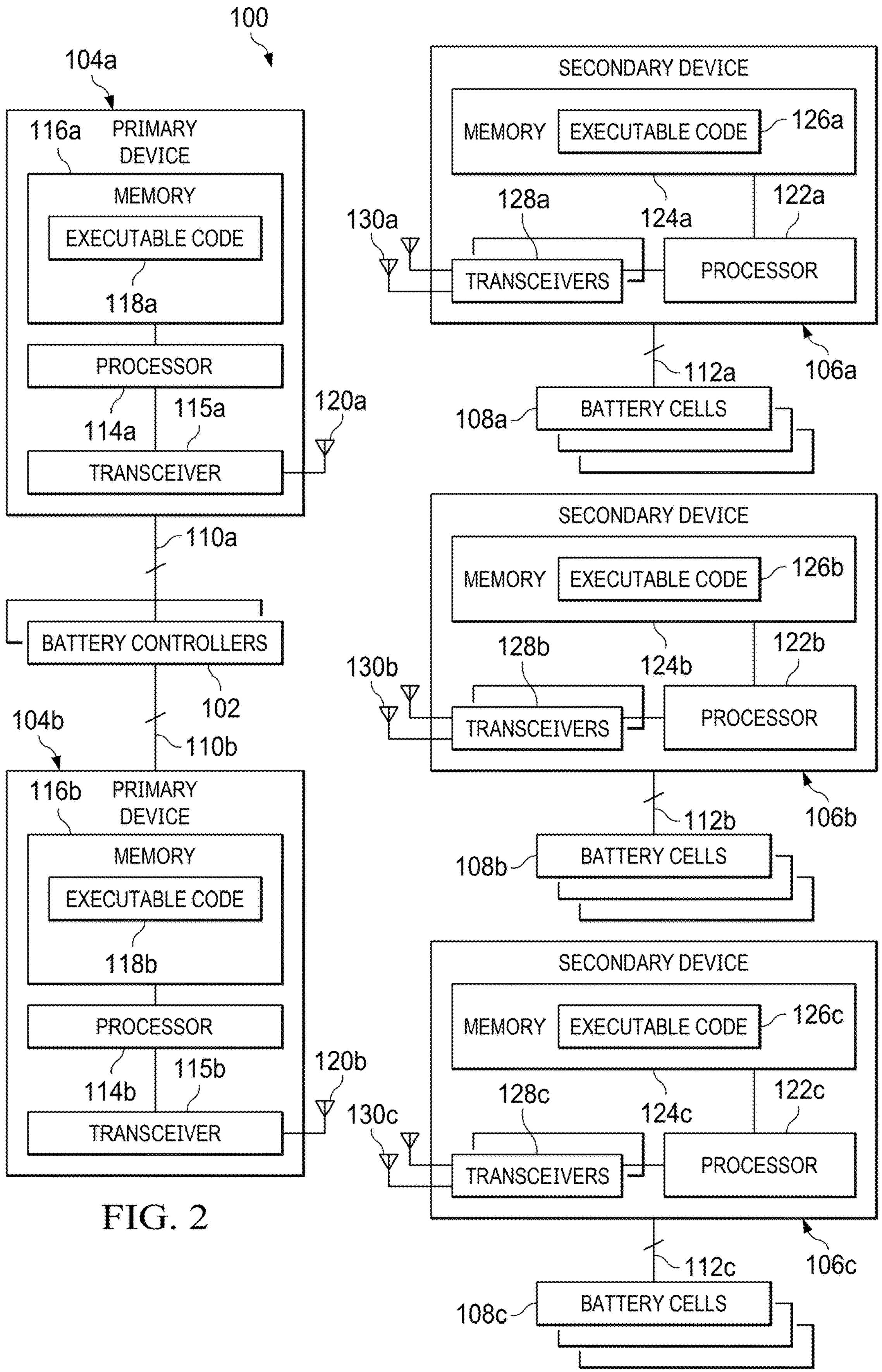
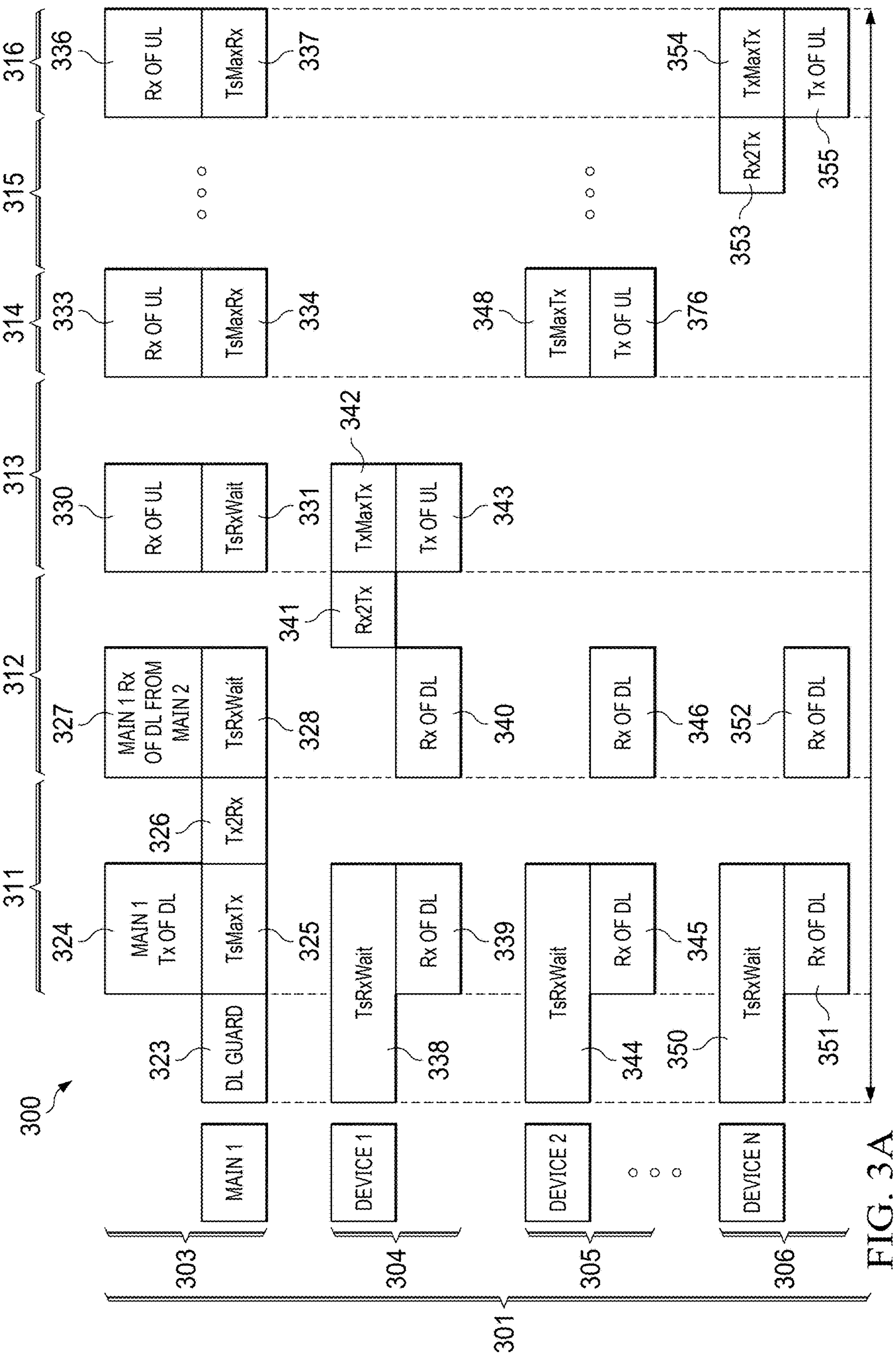

100
104a
116a
PRIMARY DEVICE
MEMORY
EXECUTABLE CODE
118a
PROCESSOR
114a
115a
TRANSCEIVER
120a
110a
BATTERY CONTROLLERS
102
104b
110b
116b
PRIMARY DEVICE
MEMORY
EXECUTABLE CODE
118b
PROCESSOR
114b
115b
TRANSCEIVER
120b
SECONDARY DEVICE
MEMORY
EXECUTABLE CODE
126a
128a
124a
122a
130a
TRANSCEIVERS
PROCESSOR
106a
112a
108a
BATTERY CELLS
SECONDARY DEVICE
MEMORY
EXECUTABLE CODE
126b
128b
124b
122b
130b
TRANSCEIVERS
PROCESSOR
106b
112b
108b
BATTERY CELLS
SECONDARY DEVICE
MEMORY
EXECUTABLE CODE
126c
128c
124c
122c
130c
TRANSCEIVERS
PROCESSOR
106c
112c
108c
BATTERY CELLS

FIG. 2


300
301
303
304
305
306
MAIN 1
DEVICE 1
DEVICE 2
DEVICE N
311
312
313
314
315
316
DL GUARD
MAIN 1 Tx OF DL
TsMaxTx
Tx2Rx
MAIN 1 Rx OF DL FROM MAIN 2
TsRxWait
Rx OF UL
TsRxWait
Rx OF UL
TsMaxRx
Rx OF UL
TsMaxRx
TsRxWait
Rx OF DL
Rx OF DL
Rx2Tx
TxMaxTx
Tx OF UL
TsMaxTx
Tx OF UL
Rx2Tx
TxMaxTx
Tx OF UL
323
324
325
326
327
328
330
331
333
334
336
337
338
339
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341
342
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344
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355
376

FIG. 3A

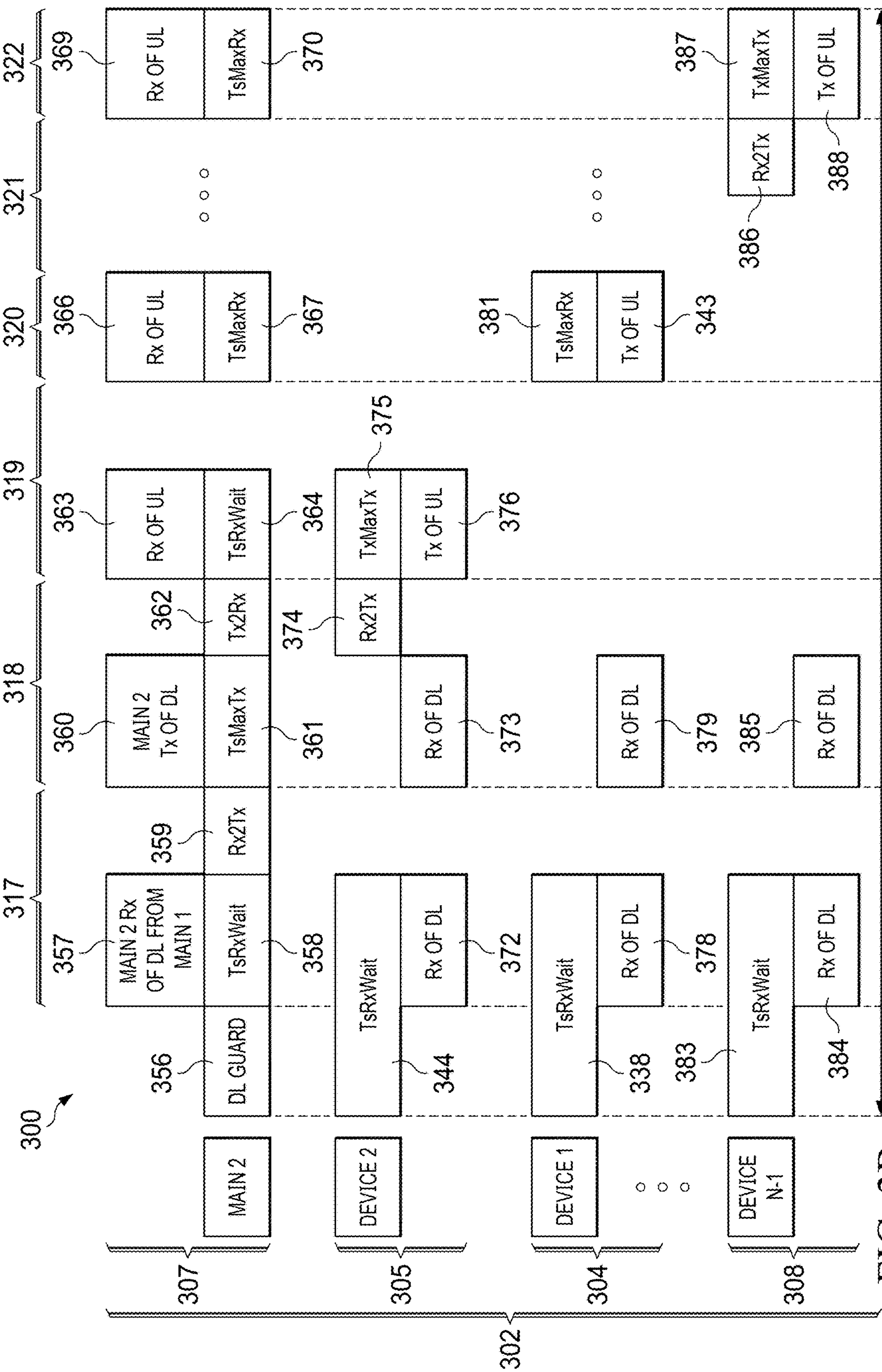
300
302
307
305
304
308
MAIN 2
DEVICE 2
DEVICE 1
DEVICE N-1
317
318
319
320
321
322
356
DL GUARD
357
MAIN 2 Rx OF DL FROM MAIN 1
358
TsRxWait
359
Rx2Tx
360
MAIN 2 Tx OF DL
361
TsMaxTx
362
Tx2Rx
363
Rx OF UL
364
TsRxWait
366
Rx OF UL
367
TsMaxRx
369
Rx OF UL
370
TsMaxRx
344
TsRxWait
372
Rx OF DL
373
Rx OF DL
374
Rx2Tx
375
TxMaxTx
376
Tx OF UL
338
TsRxWait
378
Rx OF DL
379
Rx OF DL
381
TsMaxRx
343
Tx OF UL
383
TsRxWait
384
Rx OF DL
385
Rx OF DL
386
Rx2Tx
387
TxMaxTx
388
Tx OF UL

FIG. 3B

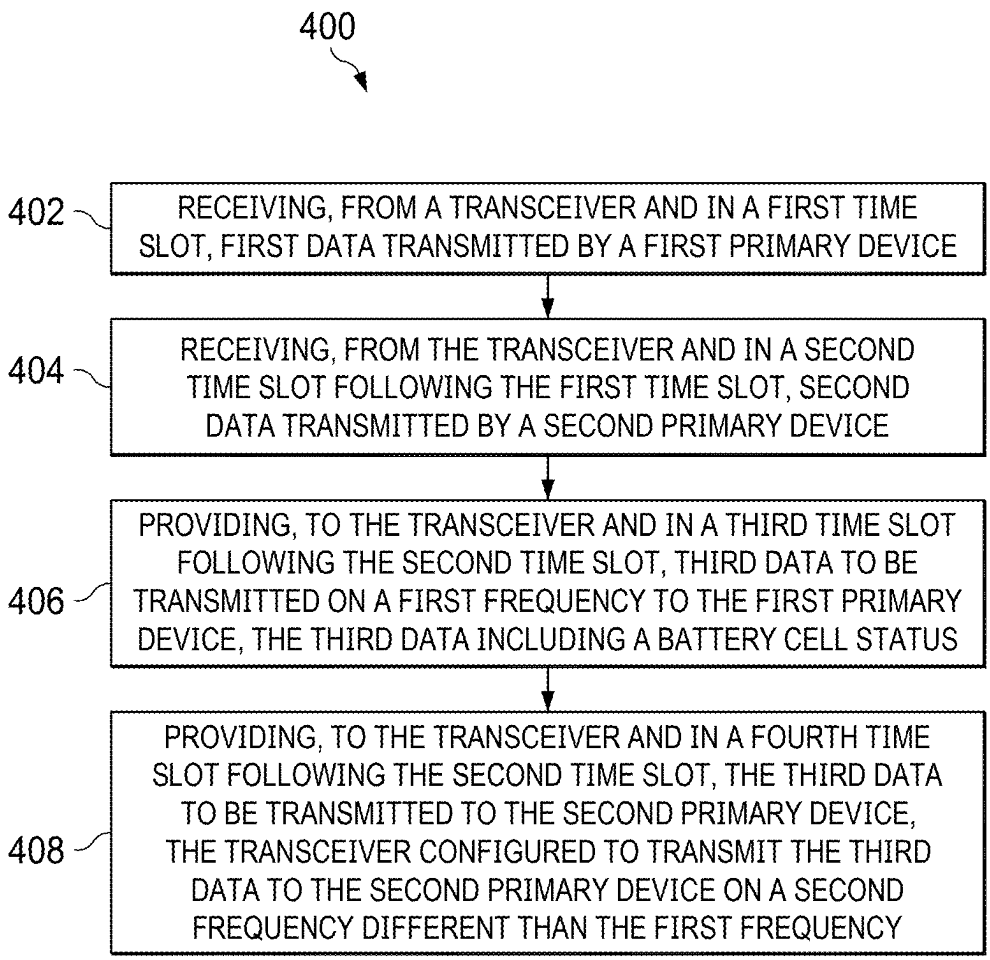
400
402
RECEIVING, FROM A TRANSCEIVER AND IN A FIRST TIME SLOT, FIRST DATA TRANSMITTED BY A FIRST PRIMARY DEVICE
404
RECEIVING, FROM THE TRANSCEIVER AND IN A SECOND TIME SLOT FOLLOWING THE FIRST TIME SLOT, SECOND DATA TRANSMITTED BY A SECOND PRIMARY DEVICE
406
PROVIDING, TO THE TRANSCEIVER AND IN A THIRD TIME SLOT FOLLOWING THE SECOND TIME SLOT, THIRD DATA TO BE TRANSMITTED ON A FIRST FREQUENCY TO THE FIRST PRIMARY DEVICE, THE THIRD DATA INCLUDING A BATTERY CELL STATUS
408
PROVIDING, TO THE TRANSCEIVER AND IN A FOURTH TIME SLOT FOLLOWING THE SECOND TIME SLOT, THE THIRD DATA TO BE TRANSMITTED TO THE SECOND PRIMARY DEVICE, THE TRANSCEIVER CONFIGURED TO TRANSMIT THE THIRD DATA TO THE SECOND PRIMARY DEVICE ON A SECOND FREQUENCY DIFFERENT THAN THE FIRST FREQUENCY

FIG. 4

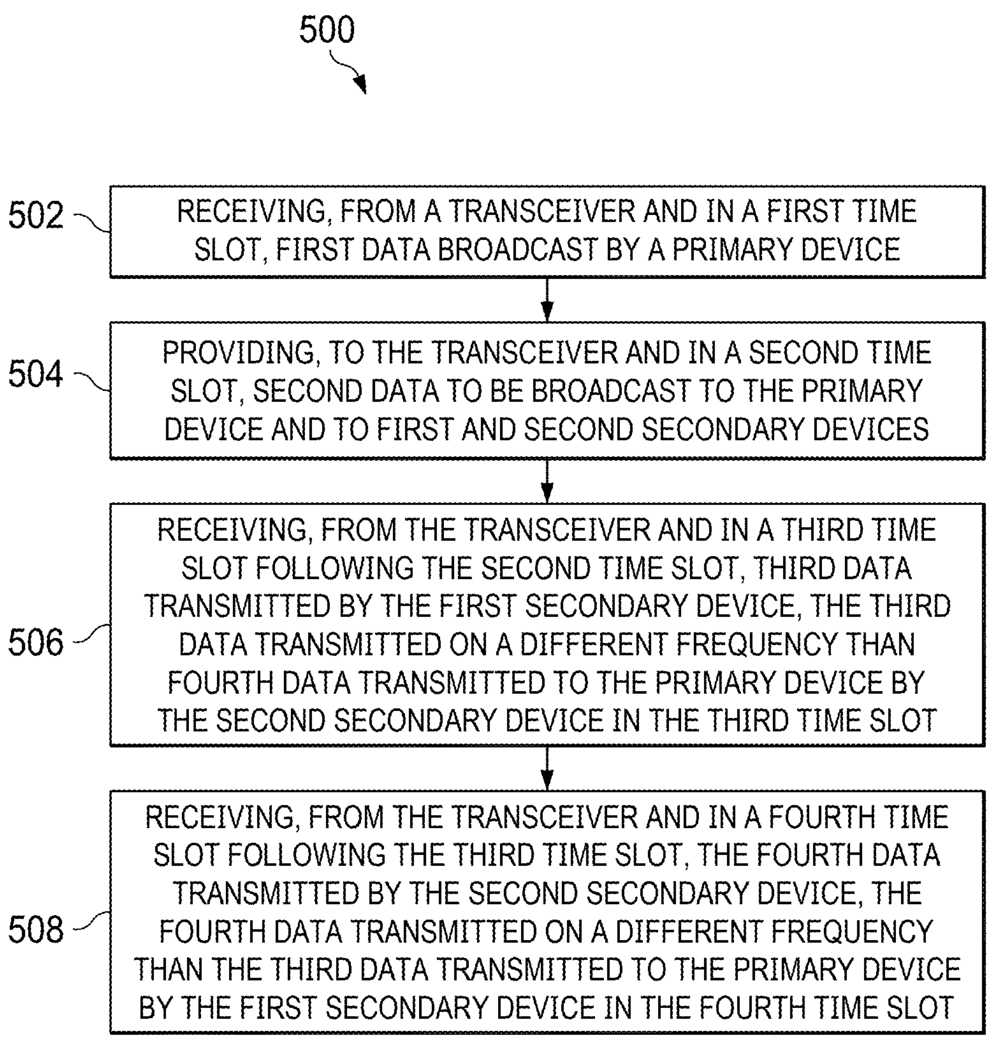
500
502
RECEIVING, FROM A TRANSCEIVER AND IN A FIRST TIME SLOT, FIRST DATA BROADCAST BY A PRIMARY DEVICE
504
PROVIDING, TO THE TRANSCEIVER AND IN A SECOND TIME SLOT, SECOND DATA TO BE BROADCAST TO THE PRIMARY DEVICE AND TO FIRST AND SECOND SECONDARY DEVICES
506
RECEIVING, FROM THE TRANSCEIVER AND IN A THIRD TIME SLOT FOLLOWING THE SECOND TIME SLOT, THIRD DATA TRANSMITTED BY THE FIRST SECONDARY DEVICE, THE THIRD DATA TRANSMITTED ON A DIFFERENT FREQUENCY THAN FOURTH DATA TRANSMITTED TO THE PRIMARY DEVICE BY THE SECOND SECONDARY DEVICE IN THE THIRD TIME SLOT
508
RECEIVING, FROM THE TRANSCEIVER AND IN A FOURTH TIME SLOT FOLLOWING THE THIRD TIME SLOT, THE FOURTH DATA TRANSMITTED BY THE SECOND SECONDARY DEVICE, THE FOURTH DATA TRANSMITTED ON A DIFFERENT FREQUENCY THAN THE THIRD DATA TRANSMITTED TO THE PRIMARY DEVICE BY THE FIRST SECONDARY DEVICE IN THE FOURTH TIME SLOT

FIG. 5

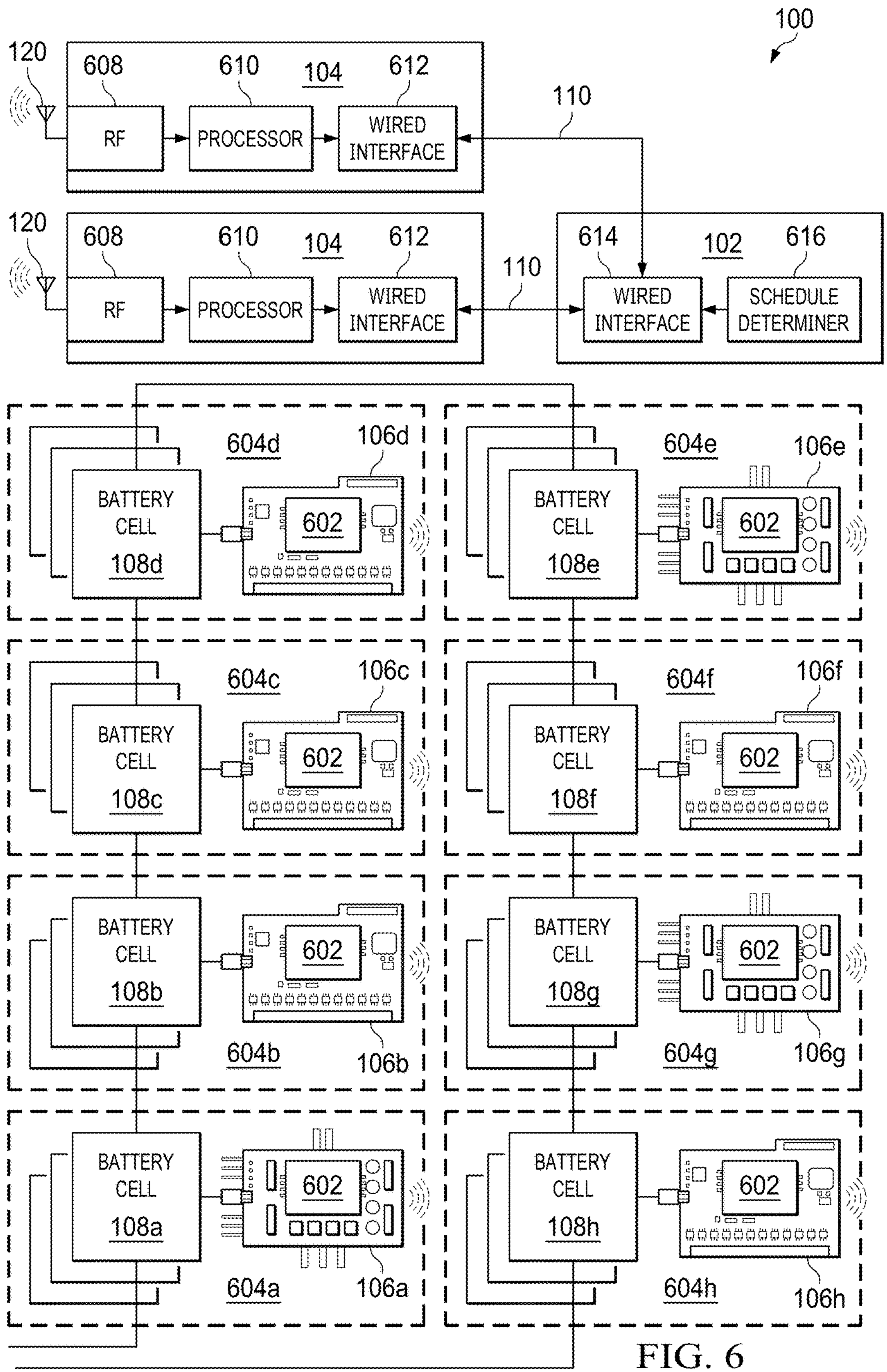
100
120
608
610
104
612
RF
PROCESSOR
WIRED INTERFACE
110
120
608
610
104
612
RF
PROCESSOR
WIRED INTERFACE
110
614
102
616
WIRED INTERFACE
SCHEDULE DETERMINER
604d
106d
BATTERY CELL 108d
602
604e
106e
BATTERY CELL 108e
602
604c
106c
BATTERY CELL 108c
602
604f
106f
BATTERY CELL 108f
602
BATTERY CELL 108b
602
604b
106b
BATTERY CELL 108g
602
604g
106g
BATTERY CELL 108a
602
604a
106a
BATTERY CELL 108h
602
604h
106h

FIG. 6

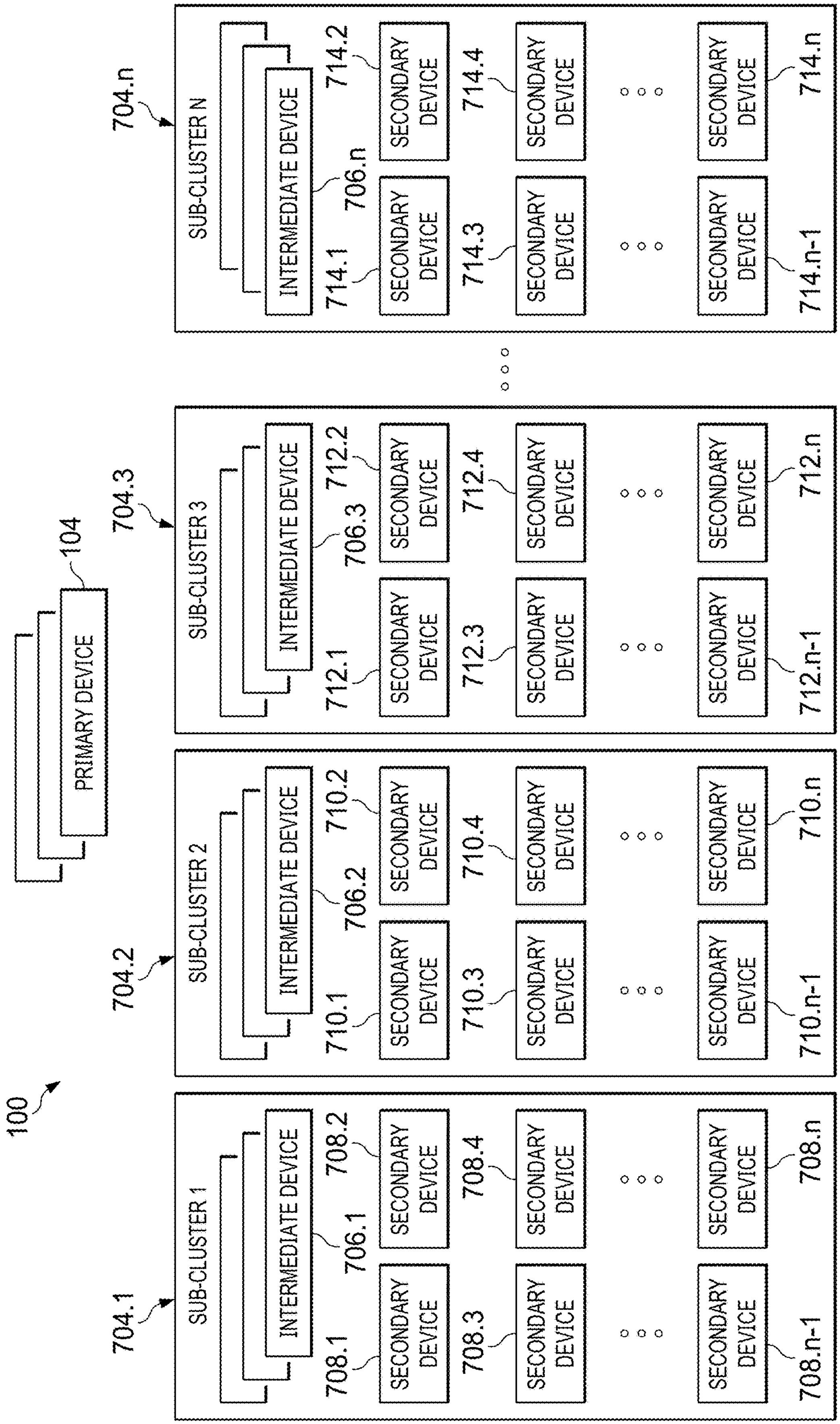
100
104
PRIMARY DEVICE
704.1
SUB-CLUSTER 1
INTERMEDIATE DEVICE
706.1
708.1
SECONDARY DEVICE
708.2
SECONDARY DEVICE
708.3
SECONDARY DEVICE
708.4
SECONDARY DEVICE
708.n-1
SECONDARY DEVICE
708.n
SECONDARY DEVICE
704.2
SUB-CLUSTER 2
INTERMEDIATE DEVICE
706.2
710.1
SECONDARY DEVICE
710.2
SECONDARY DEVICE
710.3
SECONDARY DEVICE
710.4
SECONDARY DEVICE
710.n-1
SECONDARY DEVICE
710.n
SECONDARY DEVICE
704.3
SUB-CLUSTER 3
INTERMEDIATE DEVICE
706.3
712.1
SECONDARY DEVICE
712.2
SECONDARY DEVICE
712.3
SECONDARY DEVICE
712.4
SECONDARY DEVICE
712.n-1
SECONDARY DEVICE
712.n
SECONDARY DEVICE
704.n
SUB-CLUSTER N
INTERMEDIATE DEVICE
706.n
714.1
SECONDARY DEVICE
714.2
SECONDARY DEVICE
714.3
SECONDARY DEVICE
714.4
SECONDARY DEVICE
714.n-1
SECONDARY DEVICE
714.n
SECONDARY DEVICE

FIG. 7

SCHEDULING FOR MULTIPLE PRIMARY NODES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/556,469, which was filed on Feb. 22, 2024, is entitled "SCHEDULING FOR MULTIPLE PRIMARY NODES," and is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] Modern vehicles may include multiple battery cells. Sensors may monitor information associated with the cells, such as temperature, voltage, and other indicators of cell status and health, for vehicular safety and to ensure proper operation.

SUMMARY

[0003] In examples, a device comprises a transceiver that can wirelessly transmit and receive data, and a processor coupled to the transceiver. The processor can receive, from the transceiver and in a first time slot, first data transmitted by a first primary device; receive, from the transceiver and in a second time slot following the first time slot, second data transmitted by a second primary device; provide, to the transceiver and in a third time slot following the second time slot, third data to be transmitted on a first frequency to the first primary device, the third data including a battery cell status; and provide, to the transceiver in a fourth time slot following the second time slot, the third data to be transmitted to the second primary device, the transceiver configured to transmit the third data to the second primary device on a second frequency that is different than the first frequency.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a schematic diagram of an automotive system including multiple primary and secondary devices configured to communicate using concurrent superframes, in accordance with various examples.

[0005] FIG. **2** is a block diagram of a system including multiple primary and secondary devices configured to communicate using concurrent superframes, in accordance with various examples.

[0006] FIGS. **3**A-**3**B show a timing diagram illustrating a concurrent superframe communication protocol, in accordance with various examples.

[0007] FIG. **4** is a flow diagram of a method for wireless communications using concurrent superframes, in accordance with various examples.

[0008] FIG. **5** is a flow diagram of a method for wireless communications using concurrent superframes, in accordance with various examples.

[0009] FIG. **6** is a block diagram of a system configured to communicate using concurrent superframes, in accordance with various examples.

[0010] FIG. **7** is a block diagram of a system configured to communicate using concurrent superframes, in accordance with various examples.

DETAILED DESCRIPTION

[0011] Some electronic devices operate using batteries. For example, electric vehicles include multiple battery cells that provide power to those vehicles. Because battery cells in an electronic device can provide large amounts of power, and further because the power provided by the battery cells may be vital to the operation of the electronic device, the electronic device may include a system to manage the battery cells.

[0012] Battery management systems (BMSs) may manage the battery cells of an electronic device in various ways. For example, a BMS may monitor the health (e.g., voltage, current, temperature) of battery cells in an electronic device. Further, the BMS may control various battery cells to manage the quantity of power provided by the battery cells and direct that power within the electronic device. Generally, a BMS includes multiple components, such as multiple battery modules and a controller to manage the battery modules. Each battery module, in turn, may couple to multiple battery cells and include a battery monitor to monitor those battery cells. Thus, the battery cells coupled to a battery module provide power to the electronic device; the battery monitor in the battery module monitors the health and operation of the battery cells in that battery module; and the controller communicates with the battery monitor to ensure the battery module and its cells operate properly. The controller may also communicate with the battery monitor to control the operation of the battery cells, such as to turn on, turn off, redirect, or otherwise balance the power provided by those battery cells.

[0013] BMSs may incorporate wireless technology to realize a wireless battery management system (WBMS). For example, a primary device contains or is coupled to a controller and a secondary device contains a battery module that controls multiple battery cells. The primary and secondary devices may communicate with each other wirelessly, for example using radio frequencies. In some protocols, a superframe is useful to facilitate wireless communications between the primary and secondary devices. A superframe is a wireless data communication scheme that facilitates the orderly communication of data between two or more wireless devices. In a superframe, the primary device first broadcasts a downlink (DL) communication (or packet) to multiple secondary devices. The secondary devices individually respond to the primary device with uplink (UL) communications (or packets) in a serial manner.

[0014] Systems relying on wireless communications can suffer from data losses due to a suboptimal communication environment. One reason for data losses is interference, which can arise from competing wireless signals (e.g., traffic in the same or similar frequency band), physical obstructions, or environmental conditions such as excessive moisture. Data may encounter obstacles that can degrade signal quality, leading to data loss. Additionally, wireless networks operate on shared bandwidth, meaning multiple devices contend for the same frequency spectrum, causing collisions and congestion. In some cases, multipath propagation, in which transmitted signals take multiple, different paths to reach the receiver, result in phase cancellations. These and other factors may contribute to packet loss in wireless communications.

[0015] Prior solutions have attempted to remediate such data losses by retransmitting packets in subsequent superframes. For example, a data packet intended for a primary device that was lost during transmission in a first superframe would have been retransmitted to the primary device in a second, subsequent superframe. This approach is time-consuming and inefficient. Further, this approach relies significantly on the integrity of the primary device. If the primary device ceases to function (e.g., due to power loss, malfunction, etc.), the system also ceases to function. Thus, these prior solutions are inadequate.

[0016] This disclosure describes various examples of a wireless communication technique to mitigate data losses by increasing data transmission redundancy using concurrent superframes. In particular, example systems (e.g., WBMSs) may include first and second primary devices coupled to a common battery controller. The first and second primary devices are configured for wireless communication. Example systems may further include first and second secondary devices, each of which is coupled to one or more battery cells, configured to monitor parameters of those battery cells (e.g., battery cell status), and configured to wirelessly communicate with other devices.

[0017] The primary and secondary devices described herein may be configurable to communicate wirelessly using a concurrent superframe protocol. Each superframe contains multiple time slots, and in each time slot, communications occur according to a schedule. For example, in a first time slot, device A may transmit synchronization data to device B, and in a second time slot, device B may transmit measurement data (e.g., battery status data) to device A. Superframes are generally transmitted serially, meaning one superframe after another. Superframes are not transmitted in parallel (i.e., concurrently). However, in the example systems described herein, superframes are transmitted in parallel (i.e., concurrently), and superframe slot communications are scheduled in a manner that significantly improves data redundancy and system robustness relative to systems that do not use the concurrent superframe scheme described herein.

[0018] In the concurrent superframe scheme, a first superframe and a second superframe occur concurrently, meaning that the first slot of the first superframe occurs at the same time as the first slot of the second superframe; the second slots of the first and second superframes occur concurrently, and so on. During the first slots of the first and second superframes, the aforementioned first primary device broadcasts synchronization data to the second primary device and to the first and second secondary devices. Each of the second primary device and the first and second secondary devices receives this synchronization data and uses the synchronization data to identify and store any clock differences between that device and the first primary device, and further uses such clock differences to facilitate synchronized, timely communication between the various devices. Similarly, during the second slots of the first and second superframes, the aforementioned second primary device broadcasts synchronization data to the first primary device and to the first and second secondary devices. Each of the first primary device and the first and second secondary devices receives this synchronization data and uses the synchronization data to identify and store any clock differences between that device and the second primary device, and further uses such clock differences to facilitate synchronized, timely communication between the various devices.

[0019] During the third slots of the first and second superframes (which, as described above, are concurrently occurring slots), the first secondary device transmits data to the first primary device and the second secondary device transmits data to the second primary device. The two transmissions may occur on different frequencies or channels that are adequately separated in the frequency domain to avoid interference. During the fourth and final slots of the first and second superframes (which, as described above, are concurrently occurring slots), the first secondary device transmits the same data it transmitted during the third slots, but this time, the first secondary device transmits the data to the second primary device. Similarly, during the fourth slots, the second secondary device transmits the same data it transmitted during the third slots, but this time, the second secondary device transmits the data to the first primary device. The two transmissions may occur on different frequencies or channels that are adequately separated in the frequency domain to avoid interference. In examples, all four transmissions in the third and fourth slots occur on different frequencies or channels to mitigate the risk of packet loss. By transmitting the same data twice during concurrent superframes to different primary devices on different frequencies, the likelihood that at least one instance of each datum will reach the targeted primary device (and, by extension, the battery controller coupled to the first and second primary devices) is substantially increased. In turn, the overall robustness of the system is increased. Furthermore, the use of temporally concurrent superframes increases both data throughput and performance relative to solutions in which temporally consecutive superframes are used, because, relative to consecutive superframes, the same amount of data is transmitted in half the time.

[0020] FIG. **1** is a schematic diagram of an automotive system **98** including multiple primary and secondary devices configured to communicate using concurrent superframes, in accordance with various examples. The system **98** may be a vehicle, such as an automobile, a watercraft, an aircraft, a spacecraft, or a military vehicle. The system **98** may also be a non-vehicular system that includes battery cells that are monitored. Further, the system **98** may be any type of system in which information is to be repeatedly communicated between wireless devices. Additional examples of the system **98** may include one or more of the following, in any combination: smartphones; laptop computers; desktop computers; tablets; notebooks; appliances; and entertainment devices. The remainder of this description assumes that the system **98** is an automobile, but the scope of this disclosure is not limited to automobiles or any other particular type of system. Furthermore, although the concurrent superframe scheme is described herein within the context of wireless battery management systems, the teachings of this disclosure may be extended to other types of systems, including home automation, industrial automation, and wireless sensor networks.

[0021] The example system **98** includes a wireless battery management system (WBMS) **100**. The WBMS **100** may be positioned in any part of the automobile, but in some examples, the WBMS **100** is positioned in, on, or near a bottom part of a chassis of the automobile, such as below one or more seats of the automobile. The WBMS **100** may include one or more battery controllers **102** that oversees and controls the WBMS **100** (e.g., by load balancing among devices and determining whether to measure current, voltage, temperature, or other parameters of battery cells); one or more primary devices **104** coupled to the battery controllers **102** by way of one or more wired or wireless connections **110**; one or more secondary devices **106**; and one or more battery cells **108** coupled to the one or more secondary devices **106** by way of one or more wired or wireless connections **112**. The primary devices **104** communicate wirelessly with the secondary devices **106** using a concurrent superframe protocol, as described herein.

[0022] FIG. **2** is a block diagram of the WBMS **100**, in accordance with various examples. The WBMS **100** may include one or more battery controllers **102** that are coupled to primary devices **104***a* and **104***b* by way of one or more wired or wireless connections **110***a* and **110***b*, respectively. The primary device **104***a* includes a processor **114***a*, a transceiver **115***a* coupled to the processor **114***a*, and memory **116***a* (e.g., random access memory (RAM), read-only memory (ROM)) coupled to the processor **114***a*. The memory **116***a* stores executable code **118***a*. The processor **114***a*, upon executing the executable code **118***a*, performs some or all of the actions attributed herein to the primary device **104***a* and/or to the processor **114***a*. An antenna **120***a* is coupled to the transceiver **115***a* and is configured to transmit signals to and receive signals from other devices in the WBMS **100**. The primary device **104***b* includes a processor **114***b* coupled to a transceiver **115***b* and memory **116***b* (e.g., RAM or ROM). The memory **116***b* stores executable code **118***b*. The processor **114***b*, upon executing the executable code **118***b*, performs some or all of the actions attributed herein to the primary device **104***b* and/or the processor **114***b*. An antenna **120***b* is coupled to the transceiver **115***b* and is configured to transmit signals to and receive signals from other devices in the WBMS **100**.

[0023] Any processors described herein may be implemented by any type of programmable circuitry. Examples of programmable circuitry include but are not limited to programmable microprocessors, Field Programmable Gate Arrays (FPGAs) that may instantiate instructions, Central Processor Units (CPUs), Graphics Processor Units (GPUs), Digital Signal Processors (DSPs), XPUs, or microcontrollers and integrated circuits such as Application Specific Integrated Circuits (ASICs).

[0024] Further, any transceivers described herein may use the license-free 2.4 gigahertz (GHz) industrial, scientific, and medical (ISM) band from 2.4 GHz to 2.483 GHz, which is compliant with the Bluetooth Special Interest Group (SIG). Additionally or alternatively, the transceivers may use 2 megabits per second (Mbps) Bluetooth Low Energy (BLE) across the physical layer (PHY). The Open Systems Interconnection (OSI) model includes the PHY as a layer used for communicating raw bits over a physical medium. In examples described herein, the PHY is free space, which the WBMS **100** uses to wirelessly communicate between the various devices of the WBMS **100**. In some examples, the transceivers described herein are instantiated by programmable circuitry executing RF instructions.

[0025] A secondary device **106***a* may include a processor **122***a* coupled to a memory **124***a* storing executable code **126***a*. The processor **122***a*, upon executing the executable code **126***a*, may perform some or all of the actions attributed herein to the secondary device **106***a* and/or the processor **122***a*. One or more transceivers **128***a* are coupled to the processor **122***a*, and one or more antennas **130***a* are coupled to the one or more transceivers **128***a*. The secondary device **106***a* is coupled to one or more battery cells **108***a* by way of one or more connections **112***a*.

[0026] A secondary device **106***b* may include a processor **122***b* coupled to a memory **124***b* storing executable code **126***b*. The processor **122***b*, upon executing the executable code **126***b*, may perform some or all of the actions attributed herein to the secondary device **106***b* and/or the processor **122***b*. One or more transceivers **128***b* are coupled to the processor **122***b*, and one or more antennas **130***b* are coupled to the one or more transceivers **128***b*. The secondary device **106***b* is coupled to one or more battery cells **108***b* by way of one or more connections **112***b*.

[0027] A secondary device **106***c* may include a processor **122***c* coupled to a memory **124***c* storing executable code **126***c*. The processor **122***c*, upon executing the executable code **126***c*, may perform some or all of the actions attributed herein to the secondary device **106***c* and/or the processor **122***c*. One or more transceivers **128***c* are coupled to the processor **122***c*, and one or more antennas **130***c* are coupled to the one or more transceivers **128***c*. The secondary device **106***c* is coupled to one or more battery cells **108***c* by way of one or more connections **112***c*.

[0028] The primary devices **104***a*, *b* of FIG. **2** may be collectively referred to herein as primary devices **104**. Similarly, the secondary devices **106***a-c* may be collectively referred to herein as secondary devices **106**, and so on. A similar convention may be used for any or all of the features depicted in the drawings.

[0029] The primary and secondary devices described herein (e.g., primary devices **104**, secondary devices **106**) may be implemented as a CC2662 and/or a BQ79616 made by TEXAS INSTRUMENTS INC.® of Dallas, TX. Additional example details of the CC2662 and BW79616 can be found in the datasheet entitled "CC2662R-Q1 SimpleLink™ Wireless BMS MCU," revised July 2023, available at https://www.ti.eom/product/CC2662R-Q1, and the datasheet entitled "BQ79616-Q1, BQ79614-Q1, BQ79612-Q1 Functional Safety-Compliant Automotive 16S/14S/12S Battery Monitor, Balancer and Integrated Hardware Protector," revised September 2022, available at https://www.ti.eom/product/BQ79616-Q1, each of which is incorporated by reference in its entirety.

[0030] Each of the secondary devices **106** is configured to monitor the status of respective battery cells **108** (e.g., the voltage being provided by the battery cells **108**, the temperatures of battery cells **108**). For example, each of the secondary devices **106** may include, or be coupled to, or communicate with one or more sensors configured to measure a variety of parameters associated with the battery cells **108**. The sensors may relay the sensed data to the processors **122** of the secondary devices **106** by way of a universal asynchronous receiver/transmitter (UART) protocol or any other suitable protocol. Further, each of the secondary devices **106** is configured to communicate with both of the primary devices **104**. For example, the secondary device **106***a* may communicate with the primary device **104***a* during one slot of a superframe, and the secondary device **106***a* may communicate with the primary device **104***b* during a different slot of the superframe. Alternatively, the secondary device **106***a* may communicate with the primary devices **104***a*, *b* simultaneously using different transceivers **128***a*. The secondary devices **106***b*, *c* may communicate with the primary devices **104***a*, *b* similarly to the secondary device **106***a*. In some examples, each of the secondary devices **106***a-c* is configured to communicate with one or more of the remaining secondary devices **106***a-c*. For instance, the secondary device **106***a* may be configured to communicate with one or more of the secondary devices **106***b, c*. The primary devices **104***a, b* also may communicate with the secondary devices **106***a-c* and with each other. For example, the primary device **104***a* may transmit data to one or more of the primary device **104***b,* the secondary device **106***a,* the secondary device **106***b,* or the secondary device **106***c*. The primary device **104***b* may operate similarly as the primary device **104***a*. The communications that are transmitted by the primary devices **104** may be received from one or more of the battery controllers **102**. Similarly, the communications that are received by the primary devices **104** may be provided to one or more of the battery controllers **102**.

[0031] The WBMS **100** as depicted in FIG. **2** is illustrative. In some examples, the WBMS **100** may include any number of primary devices, secondary devices, battery controllers, battery cells, other components, and wired and/or wireless connections between one or more of the foregoing. Any and all such variations are contemplated and included in the scope of this disclosure.

[0032] Additional examples of the components shown in FIGS. **1** and **2** can be found in the references incorporated by reference herein, including U.S. application Ser. No. 17/823,138, which is entitled, "Multiple Primary Nodes for Wireless Battery Management System Robustness," was filed on Aug. 30, 2022, and is hereby incorporated herein by reference in its entirety.

[0033] As described above, the various components of the WBMS **100** may communicate with each other according to a concurrent superframe protocol. A superframe interval is the time period during which a superframe occurs. Thus, "concurrent superframes" are multiple superframes that temporally overlap with each other such that they occur during a single superframe interval. FIGS. **3**A-**3**B depict a timing diagram **300** illustrating a concurrent superframe communication protocol, such as may be implemented in the WBMS **100** of FIGS. **1** and **2**, in accordance with various examples. Specifically, the timing diagram **300** includes a superframe **301** and a separate superframe **302** that occur concurrently during a single superframe interval. Each superframe includes multiple slots during which various devices in the WBMS **100** may transmit or receive data. For example, the superframe **301** includes slots **311-316**, and the superframe **302** includes slots **317-322**. Because the superframes **301**, **302** are concurrent superframes, each of the slots **311-316** occurs concurrently with a respective slot **317-322**, as shown.

[0034] Superframe **301** describes the behavior of devices **303-306**. In the example shown, device **303** may be a primary device, such as the primary devices **104***a, b*. The remaining devices **304-306** may be secondary devices, such as the secondary devices **106***a-c*. For case of description, FIGS. **3**A-**3**B label devices **303-306** as "main 1," "device 1," "device 2," and "device n," respectively. Superframe **302** describes the behavior of devices **304**, **305**, **307**, and **308**. In the example shown, device **307** may be a primary device, such as the primary devices **104***a, b*. The remaining devices **304**, **305**, and **308** may be secondary devices, such as the secondary devices **106***a-c*. For case of description, FIGS. **3**A-**3**B label devices **307**, **305**, **304**, and **308** as "main 2," "device 2," "device 1," and "device n-1," respectively.

[0035] Prior to the first slots **311** and **317**, the superframes **301** and **302** enable DL guard transmission periods **323** and **356**, respectively, which are useful to mitigate the risk of interference with other superframes that occurred prior to the superframes **301** and **302**. At the time of the DL guard transmission periods **323** and **356**, the secondary devices **304** (e.g., Node 1), **305** (e.g., Node 2), **306** (e.g., Node N), and **308** (e.g., Node N-1) enter receive modes **338**, **344**, **350**, and **383** (TsRxWait), respectively, which means these secondary devices **304**, **305**, **306**, and **308** are ready to receive transmissions from other device(s).

[0036] After the DL guard transmission periods **323** and **356** have expired, concurrent slots **311** and **317** begin. In slot **311**, the primary device **303** (e.g., Main 1) broadcasts data **324** to the remaining devices **304-308**. For example, the primary device **104***a* may broadcast data to the primary device **104***b* and the secondary devices **106***a-c* during slots **311** and **317**. The data broadcast by the primary device **303** may include synchronization information that the receiving devices may use to synchronize their clocks with a clock of the primary device **303**. For example, the data broadcast by the primary device **303** may include timestamp data that the receiving devices may use to synchronize their clocks with the clock of the primary device **303**. The data broadcast by the primary device **303** may be referred to herein as downlinks. Numeral **325** indicates that, during transmission, the primary device **303** is in a transmission mode (TsMaxTx). At the same time, the secondary devices **304**, **305**, **306**, and **308** are in receive modes **338**, **344**, **350**, and **383**, respectively (TsRxWait), and the primary device **307** (e.g., Main 2) is in a receive mode **358**. During slots **311** and **317**, the primary device **303** switches from a transmit mode to a receive mode, as numeral **326** indicates (Tx2Rx). Similarly, during slots **311** and **317**, the primary device **307** switches from a receive mode to a transmit mode, as numeral **359** indicates (Rx2Tx). The secondary devices **304**, **305**, **306**, and **308** remain in their receive modes.

[0037] Slots **312** and **318** occur concurrently following slots **311** and **317**. During slots **312** and **318**, the primary device **307** broadcasts data **360** to the remaining devices **303-306** and **308**. The broadcast data may be similar to the synchronization data described above with respect to slots **311** and **317**. During transmission of the data **360**, the primary device **307** is in a transmission mode **361** (TsMaxTx), while the primary device **303** is in a receive mode **328**. The secondary devices **304**, **305**, **306**, and **308** are also in receive modes at this time. Accordingly, the primary device **303** and secondary devices **304**, **305**, **306**, and **308** receive the data **360** broadcast by the primary device **307**. For example, primary device **104***b* may broadcast synchronization data to the primary device **104***a* and to the secondary devices **106***a-c*. After transmission and receptions are complete in slots **312** and **318**, the primary device **307** switches from a transmit mode to a receive mode as numeral **362** indicates (Tx2Rx), and secondary devices **304** and **305** switch from receive modes to transmit modes, as numerals **341** and **374** indicate (Rx2Tx).

[0038] Synchronization is complete after slots **312** and **318**. Slots **313** and **319** occur concurrently following slots **312** and **318**. During slots **313** and **319**, multiple secondary devices simultaneously transmit data (e.g., status information collected from any of the battery cells **108***a-c*) to different primary devices **303**, **307**. For example, secondary device **304** may transmit data **343** to primary device **303**, and, at the same time, secondary device **305** may transmit data **376** to primary device **307**. To enable such transmissions, secondary device **304** is in transmit mode **342** (TsMaxTx), and secondary device **305** is in transmit mode **375** (TsMaxTx). To facilitate receptions of the transmitted data, primary device **303** is in receive mode **331** (TsRxWait) and primary device **307** is in receive mode **364** (TsRxWait). The data broadcast by secondary devices to primary devices may be referred to herein as uplinks.

[0039] Slots **314** and **320** occur concurrently following slots **313** and **319**. During slots **314** and **320**, secondary device **305** again transmits the same data **376** that the secondary device **305** transmitted during slot **319**, and secondary device **304** again transmits the same data **343** that the secondary device **304** transmitted during slot **313**. However, while secondary device **305** transmits the same data **376** in both slots **319** and **314**, the destinations of the data **376** differ. During slot **319**, secondary device **305** transmits the data **376** to primary device **307**, while in slot **314**, secondary device **305** transmits the data **376** to primary device **303**. By transmitting data **376** during multiple slots and to different primary devices, secondary device **305** increases the likelihood that data **376** will be received by at least one of the primary devices **303** and **307**, and, by extension, by a battery controller (e.g., battery controllers **102**) that is coupled to primary devices **303** and **307**. In this manner, the risk of data loss is mitigated. Similarly, while secondary device **304** transmits the same data **343** in both slots **313** and **320**, the destinations of the data **343** differ. During slot **313**, secondary device **304** transmits the data **343** to primary device **303**, while in slot **320**, secondary device **304** transmits the data **343** to primary device **307**. By transmitting data **343** during multiple slots and to different primary devices, secondary device **304** increases the likelihood that data **343** will be received by at least one of the primary devices **303** and **307**, and, by extension, by a battery controller (e.g., battery controllers **102**) that is coupled to primary devices **303** and **307**.

[0040] Slots **315** and **321** occur concurrently following slots **314** and **320**. During slot **315**, secondary device **306** switches from receive mode to transmit mode, as numeral **353** indicates (Rx2Tx). Similarly, during slot **321**, secondary device **308** switches from receive mode to transmit mode, as numeral **386** indicates (Rx2Tx). Although a single slot **315** is depicted between slots **314** and **316** and a single slot **321** is depicted between slots **320** and **322**, any number of slots n may be included between slots **314** and **316**, and the same number of slots n may be included between slots **320** and **322**.

[0041] Slots **316** and **322** occur concurrently following slots **315** and **321**. During slot **316**, secondary device **306** transmits data **355** to primary device **303** (while in receive mode **354**, TsMaxTx). Although not expressly shown in FIGS. **3A-3B**, secondary device **306** will have already transmitted data **355** to primary device **307** during a prior slot between slots **314/320** and slots **316/322**. Thus, by transmitting data **355** twice during different time slots and to different primary devices, the risk of data loss is mitigated. Similarly, during slot **322**, secondary device **308** transmits data **388** to primary device **307** (while in transmit mode **387**, TsMaxTx). Although not expressly shown in FIGS. **3A-3B**, secondary device **308** will have already transmitted data **388** to primary device **303** during a prior slot between slots **314/320** and slots **316/322**. Thus, by transmitting data **388** twice during different time slots and to different primary devices, the risk of data loss is mitigated.

[0042] Redundant data transmissions need not occur in consecutive slots in all implementations of the techniques of this disclosure. For example, although timing diagram **300** depicts data **343** being transmitted in consecutive slots **313** and **320**, in examples, data **343** may be transmitted in non-consecutive slots, such as in slots **313** and **322**. (Transmission of the same data in non-consecutive time slots can be beneficial, for example, if the passage of additional time between the slots facilitates the removal of a condition or physical obstacle that was preventing successful reception of the data in the first of the two time slots.) The data transmission redundancy contemplated herein may be achieved so long as a particular datum is transmitted twice in a given superframe interval, to differing primary devices, and on different frequencies, meaning that the datum is transmitted to a first primary device, on a first frequency, and in a first superframe, and then again to a second primary device, on a second frequency different from the first frequency, and in a second superframe concurrent with the first superframe. Data redundancy also may be achieved even if the datum is transmitted twice to the same primary device. For example, a particular datum may be transmitted to a first primary device, on a first frequency, and in a first superframe, and then again to the same first primary device, on a second frequency different than the first frequency, and in a second superframe concurrent with the first superframe. Data redundancy also may be achieved even if the datum is transmitted twice on the same frequency. For example, a particular datum may be transmitted to a first primary device, on a first frequency, and in a first superframe, and then again to a second primary device, on the same first frequency, and in a second superframe that is concurrent with the first superframe. Data redundancy also may be achieved even if the datum is transmitted twice on the same frequency and to the same primary device. For example, a particular datum may be transmitted to a first primary device, on a first frequency, and in a first superframe, and then again to the same first primary device, on the same first frequency, but in a second superframe concurrent with the first superframe. Any and all such variations are contemplated and included in the scope of this disclosure.

[0043] To mitigate the risk of interference between simultaneous data transmissions, and further to mitigate the risk that same-data transmissions during separate slots fail to reach their intended destinations, secondary devices may use any of a variety of frequency hopping schemes. For example, with reference to timing diagram **300**, in concurrent slots **313** and **319**, the data **343** may be transmitted on a first frequency, and the data **376** may be simultaneously transmitted on a second frequency that is different than the first frequency. The first and second frequencies may be separated by at least 5 MHz or by at least 10 MHz (or, in terms of channels, the channels used are at least two channels, three channels, or five channels apart). Similarly, in concurrent slots **314** and **320**, the data **343** and **376** may be simultaneously transmitted on first and second frequencies, respectively, with the first and second frequencies separated by at least 5 MHz or by at least 10 MHz (or, in terms of channels, the channels used are at least two channels, three channels, or five channels apart). Using different frequencies for simultaneous data transmission mitigates the risk of interference between the transmissions.

Separation of at least 5 MHz or 10 MHz are examples of minimum separations; in some examples, the techniques of this disclosure may be implemented with minimum frequency separation other than 5 MHz or 10 MHz.

[0044] Different frequencies also may be useful across different slots. For example, in timing diagram **300**, the data **343** may be transmitted on a first frequency in slot **313** and on a second frequency in slot **320** that is separated from the first frequency by at least 5 MHz or by at least 10 MHz (or, in terms of channels, the channels used are at least two channels, three channels, or five channels apart). Similarly, the data **376** may be transmitted on a first frequency in slot **319** and on a second frequency in slot **314** that is separated from the first frequency by at least 5 MHz or by at least 10 MHz (or, in terms of channels, the channels used are at least two channels, three channels, or five channels apart). To facilitate such separation in transmission frequencies, primary devices (e.g., primary devices **104**) may transmit in their downlinks to secondary devices (e.g., secondary devices **106**) specific frequency hopping schemes that are to be used during data transmissions. Furthermore, in some examples, secondary devices may receive and transmit data on similar frequencies. For example, with reference to timing diagram **300**, the secondary device **304** may receive data **324** in slot **311** on a first frequency, receive data **360** in slot **318** on a second frequency, transmit data **343** in slot **313** on a third frequency, and transmit data **343** in slot **320** on a fourth frequency. In examples, the first and third frequencies may be the same. In examples, the first and fourth frequencies may be the same. In examples, the second and third frequencies may be the same. In examples, the second and fourth frequencies may be the same.

[0045] As just one example, a primary device can use a frequency hopping sequence with thirty-seven channels in the industrial, scientific, and medical (ISM) band. These bands have a random order in the hopping sequence and satisfy the condition that the adjacent frequencies are separated by 5 MHz or by 10 MHz. For example, the first hopping sequence is {2410, 2404, 2416, . . . }. The second hopping sequence can be a different version of the first hopping sequence, except shifted by one position. Hence, the second hopping sequence is {2404, 2416, . . . 2410}, preserving a 5 MHz separation or a 10 MHz separation for the primary device operating in a first frequency (2410) and another primary device operating in a second frequency (2404). After each hop, the first and second frequencies remain 5 MHz apart or 10 MHz apart.

[0046] In addition to transmitting frequency hopping schemes in downlinks, primary devices (e.g., primary devices **104**) may transmit scheduling instructions to secondary devices (e.g., secondary devices **106**) in downlinks. For example, one or more primary devices may be configured to instruct secondary devices regarding the specific slots of a given superframe(s) and the specific channels and/or frequencies in which the secondary devices are to transmit uplinks to the primary devices. Primary devices may change the scheduling instructions with each superframe interval, or, alternatively, may maintain the same scheduling instructions for multiple consecutive superframe intervals. Primary devices may also transmit additional information to the secondary devices, such as acknowledgements for uplink transmissions from a previous superframe, an indication when the next superframe may begin, an adaptive frequency hopping countdown, etc. In some examples, the primary devices do not transmit scheduling instructions to the secondary devices in the downlinks. Instead, some or all devices in the WBMS **100** are preprogrammed with a defined schedule that is to be followed for some or all superframe intervals, unless instructed otherwise by one or more primary devices. Alternatively, primary devices may transmit a single downlink with scheduling instructions that are to be followed in all superframe intervals until further notice. A battery controller (e.g., battery controller **102**) may provide the primary devices with scheduling instructions that the primary devices may then disseminate to the remaining devices of the WBMS **100**.

[0047] The secondary devices transmit data to the primary devices according to the scheduling instructions and channel and/or frequency instructions provided by the primary devices or preprogrammed into the secondary devices. After receiving a downlink transmission from a primary device, a secondary device may parse the transmission to determine the slots and channels or frequencies in which the secondary device is scheduled to uplink information to the primary device(s). In some examples, information for how to parse the downlink transmission may be provided to the secondary device during a WBMS network formation process.

[0048] In some examples, secondary devices include multiple transceivers. For example, as shown in FIG. **2**, secondary device **106*a*** includes multiple transceivers **128*a*.** In examples, the multiple transceivers **128*a*** of FIG. **2** may simultaneously (in the same slot) transmit data **343** twice using different transceivers **128*a*.** Such simultaneous transmissions may occur on the same or different frequencies. The transmissions may be directed to the same or different primary devices **104** (e.g., both the secondary and primary devices may have multiple transceivers communicating simultaneously on different frequencies). Thus, the same level of redundancy that is achieved by single-transceiver secondary devices transmitting the same data in different slots can be achieved by multi-transceiver secondary devices in a more compressed timeframe.

[0049] In some examples, primary devices (e.g., primary devices **104***a, b*) may be spatially positioned apart from each other (e.g., 6.25 cm (half the wavelength at 2.4 GHz) apart). Providing a threshold amount of distance between the primary devices increases the likelihood that environmental obstacles hindering the successful transmission of data packets to one primary device will not likewise hinder the successful transmission of data packets to the other primary device.

[0050] In some examples, uplink data transmissions may be performed in consecutive slots of concurrent superframes to accommodate large amounts of data that could not otherwise be transmitted in a single uplink. Such techniques are described in U.S. patent application Ser. No. 18/227,821, which is entitled "Methods and Apparatus to Determine Communication Schedules for Wireless Battery Systems," was filed on Jul. 28, 2023, and is hereby incorporated herein by reference in its entirety. All subject matter described in U.S. patent application Ser. No. 18/345,636, which is entitled, "Hierarchical Wireless Battery Management System" and was filed on Jun. 30, 2023, is hereby incorporated herein by reference in its entirety.

[0051] FIG. **4** is a flow diagram of a method **400** for wireless communications using concurrent superframes, in accordance with various examples. In examples, a secondary device (e.g., secondary device **106***a, b,* or *c*) performs the method **400** using the example concurrent superframe scheme depicted in FIGS. **3A-3B**. Thus, the method **400** is described herein from the perspective of example secondary device **106***a* and with reference to FIGS. **2-4**.

[0052] The method **400** includes receiving, from a transceiver and in a first time slot, first data transmitted by a first primary device (**402**). For example, the processor **122***a* of secondary device **106***a* may receive, by way of a transceiver **128***a* and during slot **311** (which occurs concurrently with slot **317**), first data transmitted by the primary device **104***a*. This data may, for instance, be the data **324**, which, as described above, is broadcast by the primary device **303** (such as primary device **104***a*) to the remaining primary and secondary devices **304-308** (which may include secondary device **106***a*).

[0053] The method **400** includes receiving, from the transceiver and in a second time slot following the first time slot, second data transmitted by a second primary device (**404**). For example, the processor **122***a* of secondary device **106***a* may receive, by way of a transceiver **128***a* and during slot **312** (which occurs concurrently with slot **317**), second data transmitted by the primary device **104***b*. This data may, for instance, be the data **360**, which, as described above, is broadcast by the primary device **307** (such as primary device **104***b*) to the remaining primary and secondary devices **303-306** and **308** (which may include secondary device **106***a*).

[0054] The method **400** includes providing, to the transceiver and in a third time slot following the second time slot, third data to be transmitted on a first frequency to the first primary device, the third data including a battery cell status (**406**). For example, the processor **122***a* of secondary device **106***a* may transmit, by way of a transceiver **128***a* and during slot **313** (which occurs concurrently with slot **319**), third data to the first primary device. This data may, for instance, be the data **343**, which, as described above, is transmitted by secondary device **304** (such as secondary device **106***a*) to primary device **303** (such as primary device **104***a*). The data may include battery cell status, such as status information (e.g., voltage, current, temperature) pertaining to the battery cells **108***a*. The transceiver **128***a* may transmit the third data on a particular frequency that differs from the frequency on which other data transmitted during the third slot (e.g., slot **313**) in transmitted. In examples, the difference in frequencies is at least 5 MHz or at least 10 MHz.

[0055] The method **400** includes providing, to the transceiver and in a fourth time slot following the second time slot, the third data to be transmitted to the second primary device, the transceiver configured to transmit the third data to the second primary device on a second frequency different than the first frequency (**408**). For example, the processor **122***a* of secondary device **106***a* may transmit, by way of a transceiver **128***a* and during slot **314** (which occurs concurrently with slot **320**), the same third data that was transmitted in the third time slot of step **406**. However, in the fourth slot of step **408**, the third data (e.g., data **343**) is transmitted to a different primary device (e.g., primary device **307**, such as primary device **104***b*) and on a different frequency than the frequency used in step **406** (e.g., with the transmission frequencies in steps **406** and **408** being separated by at least 5 MHz or by at least 10 MHz).

[0056] By transmitting the same data on different frequencies and in different slots, secondary device **106***a* raises the likelihood that the data will reach the intended destination. The risk of data loss is mitigated.

[0057] FIG. **5** is a flow diagram of a method **500** for wireless communications using concurrent superframes, in accordance with various examples. In examples, a primary device (e.g., primary device **104***a* or b) performs the method **500** using the example concurrent superframe scheme depicted in FIGS. **3A-3B**. Thus, the method **500** is described herein from the perspective of example primary device **104***b*.

[0058] The method **500** includes receiving, from a transceiver and in a first time slot, first data broadcast by a primary device (**502**). For example, the processor **114***b* of primary device **104***b* (e.g., primary device **307**) may receive, from the transceiver **115***b* and in slot **311**, first data (e.g., data **324**) broadcast by primary device **104***a* (e.g., primary device **303**).

[0059] The method **500** includes providing, to the transceiver and in a second time slot, second data to be broadcast to the primary device and to first and second secondary devices (**504**). For example, the processor **114***b* of primary device **104***b* (e.g., primary device **307**) may provide to the transceiver **115***b* and in slot **313**, second data (e.g., data **360**) to be broadcast to the primary device (e.g., primary device **303**, **104***a*) and to first and second secondary devices (e.g., secondary devices **304**, **305**, such as secondary devices **106***a, b*).

[0060] The method **500** includes receiving, from the transceiver and in a third time slot following the second time slot, third data transmitted by the first secondary device, the third data transmitted on a different frequency than fourth data transmitted to the primary device by the second secondary device in the third time slot (**506**). For example, the processor **114***b* of primary device **104***b* (e.g., primary device **307**) may receive, from the transceiver **115***b* and in slot **313**, third data (e.g., data **376**) transmitted on a different frequency than fourth data (e.g., data **343**) transmitted to the primary device **303** (e.g., primary device **104***a*) by the second secondary device (e.g., secondary device **304**, **106***b*). The data **376** may have been transmitted by secondary device **305** (e.g., secondary device **106***a*). The frequencies on which data **343** and **376** are transmitted in the third time slot are separated by at least 5 MHz or by at least 10 MHz.

[0061] The method **500** includes receiving, from the transceiver and in a fourth time slot following the third time slot, the fourth data transmitted by the second secondary device, the fourth data transmitted on a different frequency than the third data transmitted to the primary device by the first secondary device in the fourth time slot (**508**). For example, the processor **114***b* of primary device **104***b* (e.g., primary device **307**) may receive, from the transceiver **115***b* and in slot **314**, the fourth data (e.g., data **343**) transmitted by the second secondary device (e.g., secondary device **304**, **106***b*). The third data (e.g., data **376**) is also transmitted during slot **314** to secondary device **305** (e.g., secondary device **106***a*). The third and fourth data are transmitted on differing frequencies during slot **314**, with the frequencies separated by at least 5 MHz or by at least 10 MHz.

[0062] In this manner, data **343** is transmitted twice (e.g., once during slot **313**, and once during slot **320**), and data **376** is transmitted twice (e.g., once during slot **319**, and once during slot **314**). By transmitting each data twice in different slots of different, concurrent superframes (i.e., twice during the same superframe interval) and on different frequencies, the likelihood of at least one of the two data transmissions reaching the battery controllers **102** is significantly increased.

[0063] FIG. **6** is a block diagram of an example implementation of the WBMS **100** of FIGS. **1** and **2**. The WBMS **100** of FIG. **6** is configured to operate in accordance with the concurrent superframe scheme described herein, such as that described with reference to FIGS. **3-5**. The WBMS **100** of FIG. **6** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by programmable circuitry such as a Central Processor Unit (CPU) executing first instructions. Additionally or alternatively, the WBMS **100** of FIG. **6** may be instantiated by (i) an Application Specific Integrated Circuit (ASIC) and/or (ii) a Field Programmable Gate Array (FPGA) structured and/or configured in response to execution of second instructions to perform operations corresponding to the first instructions. Some or all of the circuitry of FIG. **6** may, thus, be instantiated at the same or different times. Some or all of the circuitry of FIG. **6** may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. **6** may be implemented by microprocessor circuitry executing instructions and/or FPGA circuitry performing operations to implement one or more virtual machines and/or containers.

[0064] The example block diagram of FIG. **6** includes the battery controller **102**, primary device **104** (e.g., primary device **104***a* or **104***b*), secondary devices **106***a*-**106***h* (which correspond to secondary devices **106**), and battery cells **108***a*-**108***h* (which correspond to battery cells **108**). The scope of this disclosure is not limited to any particular number of each type of device illustrated in FIG. **6**. For example, while two instances of primary device **104** are expressly shown, the WBMS **100** of FIG. **6** may include any number of primary devices **104**.

[0065] Each of the secondary devices **106** includes example schedule requester circuitry **202**. As used herein, a secondary device **106***a* and its corresponding battery cell **108***a* are collectively referred to as a battery module **204***a*. Accordingly, the battery modules **204***a*-**204***h* collectively form battery modules **204**. The primary device **104** includes an example antenna **206**, example radio frequency (RF) transceiver **208**, an example processor **210** and example wired interface circuitry **212**. The battery controller **102** includes example wired interface circuitry **214** and example schedule determiner circuitry **216**. While FIG. **6** shows eight battery modules **204** and one battery controller **102**, in other examples, the WBMS **100** includes any number of battery modules and battery controllers. For example, WBMS **100** may include two or more battery controllers, where a first set of the secondary devices are assigned to communicate with a first battery controller, and a second set of the secondary devices are assigned to communicate with a second battery controller. The first and second sets of secondary devices may or may not overlap.

[0066] The battery modules **204** wirelessly communicate with the primary device **104** using concurrent superframes. That is, a first superframe containing a first set of communications occurs at the same time as a second superframe containing a second set of communications. Within a given battery module **204***a,* the schedule requester circuitry **202** determines whether a transmission regarding the corresponding battery cell **108***a* should be made in an upcoming set of concurrent superframes. The schedule requester circuitry **202** may determine whether to make a transmission based on factors that include status and performance of the corresponding battery cell **108***a*.

[0067] The schedule requester circuitry **202** optionally requests to be included on a schedule for an upcoming superframe based on the result of the determination and in accordance with the teachings of this disclosure. Accordingly, the battery modules **204** do not request to make a transmission in an upcoming superframe every time an opportunity to make a request is available. The schedule requester circuitry **202** may provide additional information to the battery controller **102** when requesting a transmission in an upcoming superframe. The schedule requester circuitry **202** may also request a specific number of requested time slots, request a specific duration of uplink time, and/or request a specific data size to uplink (e.g., a specific number of blocks, bytes, or bits), etc. Alternatively, the request sent by the schedule requester circuitry **202** may indicate only that a corresponding battery module **204**A is requesting more time for transmission, without any specifics about the requested time duration, number of time slots, or uplink size.

[0068] In the example of FIG. **6**, each instance of the schedule requester circuitry **202** is implemented within the secondary devices **106**. In other examples, one or more instances of the schedule requester circuitry **202** are implemented elsewhere within the respective battery modules **204**. In some examples, the schedule requester circuitry **202** is instantiated by programmable circuitry executing schedule requester instructions.

[0069] Within the primary device **104**, the radio frequency (RF) transceiver **208** communicates wirelessly with the secondary devices **106** via the antenna **206**. The RF transceiver **208** may use the license-free 2.4 gigahertz (GHz) industrial, scientific, and medical (ISM) band from 2.4 GHz to 2.483 GHz, which is compliant with the Bluetooth Special Interest Group (SIG). Additionally or alternatively, the RF transceiver **208** may use 2 megabits per second (Mbps) Bluetooth Low Energy (BLE) across the physical layer (PHY). The Open Systems Interconnection (OSI) model includes the PHY as a layer used for communicating raw bits over a physical medium. In examples described herein, the PHY is free space, which the wireless battery management system **100** uses to wirelessly communicate between the primary device **104** and the secondary devices **106**. In some examples, the RF transceiver **208** is instantiated by programmable circuitry executing RF instructions. The remaining transceivers in WBMS **100** may be configured to operate similarly as the RF transceiver **208**.

[0070] Within the primary device **104**, the processor **210** both interprets the contents of data received by the primary device **104** and determines the contents of data to be transmitted by the primary device **104**. In doing so, the processor **210** helps establish communication between the battery modules **204** and the battery controller **102**. The processor **210** may be implemented by any type of programmable circuitry. Examples of programmable circuitry include but are not limited to programmable microprocessors, Field Programmable Gate Arrays (FPGAs) that may instantiate instructions, Central Processor Units (CPUs), Graphics Processor Units (GPUs), Digital Signal Processors (DSPs), XPUs, or microcontrollers and integrated circuits such as Application Specific Integrated Circuits (ASICs).

[0071] Within the primary device **104**, the wired interface circuitry **212** sends and receives communications with the battery controller **102** via the wired connection **110**. The wired interface circuitry **212** may implement any suitable hardware components, including but not limited to terminals, pins, interconnects, etc., to implement wired communications. Similarly, within the battery controller **102**, the wired interface circuitry **214** sends and receives communications with the primary device **104** via the wired connection **110**. The wired interface circuitry **214** may implement any suitable hardware components to implement wired communications. In examples, the wired interfaces **212**, **214** may be replaced by transceivers or other circuitry suitable to facilitate wireless communications between the primary device **104** and the battery controller **102**.

[0072] The schedule determiner circuitry **216** determines different communication schedules for different superframes. The schedule determiner circuitry **216** adjusts a schedule and/or creates new schedules for superframes based on the transmission request transmitted by the multiple instances of the schedule requester circuitry **202**.

[0073] In the example of FIG. **6**, the schedule determiner circuitry **216** is implemented within the battery controller **102**. In other examples, the schedule determiner circuitry **216** is implemented within the primary device **104** or elsewhere within the WBMS **100**. In some examples, the schedule determiner circuitry **216** is instantiated by programmable circuitry executing schedule determiner instructions.

[0074] The battery modules **204** are heterogeneous in the sense that the design, manufacture, capabilities, and/or performance of a first battery module may differ from that of a second battery module. For example, in FIG. **6**, battery cells **108***a*, **108***b*, **108***d* store a larger amount of charge than battery cells **108***c*, **108***e*-**108***h*. Furthermore, the amount of charge stored in battery cells **108***a*, **108***b*, **108***d* is nonuniform. In an additional example, FIG. **6** illustrates the secondary devices **106***a*, **106***e*, **106***g* implemented by a first type of programmable circuitry, and secondary devices **106***b*-**106***d*, **106***f*, **106***h* implemented by a different type of programmable circuitry. While the example FIG. **6** illustrates variance in battery capacity and type of programmable circuitry, in practice, the battery modules **204** may include other types of differences.

[0075] In some examples, the heterogeneity of the WBMS **100** causes some battery modules to seek communication with the battery controller **102** more frequently than other battery modules. Some battery modules may additionally or alternatively transmit different types of information within a superframe than other battery modules. For example, the battery module **204***a* may seek to report a storage capacity measurement when the battery module **204***b* seeks to report an error code. The battery controller **102** enables such diverse forms of communication by obtaining requests for transmissions sent by the battery modules **204** and determining a schedule for each superframe.

[0076] FIG. **7** is a block diagram of another example WBMS **100**. The architecture of the example WBMS **100** in FIG. **7** includes a layer of intermediate devices between the primary and secondary devices, thus facilitating scale in especially large or complex systems. The example WBMS **100** of FIG. **7** includes one or more primary devices **104** and multiple sub-clusters **704.1**-**704**.N (collectively referred to herein as sub-clusters **704**). Although FIG. **7** shows an example WBMS **100** with a single primary device **104**, the sub-cluster architecture shown in FIG. **7** may be implemented in a WBMS **100** with multiple primary devices (e.g., WBMS **100** shown in FIG. **2**). The WBMS **100** shown in FIG. **7** may be configurable to implement the scheduling described herein with respect to FIGS. **3**-**5** with the intermediate devices **706** communicating with multiple primary devices.

[0077] The sub-cluster **704.1** includes one or more intermediate devices **706.1** and multiple secondary devices **708.1**-**708**.N (collectively referred to herein as secondary devices **708**). The sub-cluster **704.2** includes one or more intermediate devices **706.2** and multiple secondary devices **710.1**-**710**.N (collectively referred to herein as secondary devices **710**). The sub-cluster **704.3** includes one or more intermediate devices **706.3** and multiple secondary devices **712.1**-**712**.N (collectively referred to herein as secondary devices **712**). The sub-cluster **704**.N includes one or more intermediate devices **706**.N and multiple secondary devices **714.1**-**714**.N (collectively referred to herein as secondary devices **714**).

[0078] In operation, each of the secondary devices shown in FIG. **7** collects data, such as the battery cell status data described above, and provides the data to a corresponding intermediate device **706**. For example, each of the secondary devices **708** collects data and transmits the data to first and second intermediate devices **706.1**. These transmissions occur according to the concurrent superframe scheme described herein, such as with reference to FIGS. **1**-**5**. Thus, for instance, the first and second intermediate devices **706.1** may broadcast downlink synchronization information during first and second slots of concurrent superframes, as described above. The secondary devices **708** may receive the broadcast information and use the broadcast information to synchronize communications with the first and second intermediate devices **706.1**. Thereafter, during the concurrent superframes, each of the secondary devices **708** may transmit its respective data twice, once in one slot to the first intermediate device **706.1**, and again in another slot to the second intermediate device **706.1**, with both transmissions occurring on different frequencies. In this way, the first and second intermediate devices **706.1** are substantially likely to receive at least one instance of the data from each of the secondary devices **708**. Each of the sub-clusters **704** operates in a similar manner.

[0079] After the intermediate devices **706** of each sub-cluster **704** has received the data from respective secondary devices, the intermediate devices **706** transmit the data to the primary devices **104** using the concurrent superframe scheme described herein. For example, in first and second slots of concurrent superframes, first and second primary devices **104** may broadcast downlink synchronization information to the intermediate devices **706**, which the intermediate devices **706** may use to synchronize communications with the first and second primary devices **104**. Thereafter, during the concurrent superframes, each of the intermediate devices **706** may transmit respective data twice, once in one slot to the first primary device **104**, and again in another slot to the second primary device **104**, with both transmissions possibly occurring on different frequencies. In this way, the first and second primary devices **104** are substantially likely to receive at least one instance of the data from each of the intermediate devices **706**. The concurrent superframe scheme described herein may be scaled to any number of wireless devices in a WBMS, or any other system besides a WBMS in which robust wireless communications are useful. For example, three or more primary devices **104** may be used, in which case a given intermediate device **706** may transmit the same data to the three or more primary devices **104** during different slots and on different frequencies. In some examples in which three or more primary devices **104** may be used, a first intermediate device **706** may transmit data to first and second primary devices **104** in different slots and on different frequencies, while a second intermediate device **706** transmits different data to second and third primary devices **104** in different slots and on different frequencies. Furthermore, the concepts described herein may be extended to any number of concurrent superframes.

[0080] In this description, the term "couple" may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0081] A device that is "configured to" perform a task or function may be configured (e.g., programmed and/or hard-wired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0082] In this description, unless otherwise stated, "about," "approximately" or "substantially" preceding a parameter means being within +/−10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

What is claimed is:

1. A device, comprising:

a transceiver configured to wirelessly transmit and wirelessly receive data; and

a processor coupled to the transceiver, the processor configured to:

receive, from the transceiver and in a first time slot, first data transmitted by a first primary device;

receive, from the transceiver and in a second time slot following the first time slot, second data transmitted by a second primary device;

provide, to the transceiver and in a third time slot following the second time slot, third data to be transmitted on a first frequency to the first primary device, the third data including a battery cell status; and

provide, to the transceiver and in a fourth time slot following the second time slot, the third data to be transmitted to the second primary device, the transceiver configured to transmit the third data to the second primary device on a second frequency that is different than the first frequency.

2. The device of claim **1**, wherein the third and fourth time slots are consecutive time slots.

3. The device of claim **1**, wherein the third and fourth time slots are non-consecutive time slots.

4. The device of claim **1**, wherein the first data or the second data specifies a timing of the third time slot or the fourth time slot.

5. The device of claim **1**, wherein the device is configured to store a timing of the third time slot or the fourth time slot prior to receipt of the first data and prior to receipt of the second data.

6. The device of claim **1**, wherein the first and second frequencies are separated by at least 5 MHz.

7. The device of claim **1**, wherein the first and second time slots are consecutive time slots.

8. The device of claim **1**, wherein the processor is configured to receive the first data in the first time slot on a third frequency that is the same as the first frequency.

9. The device of claim **1**, wherein the processor is configured to receive the second data in the second time slot on a third frequency that is the same as the second frequency.

10. The device of claim **1**, wherein the processor is configured to receive the first data in the first time slot on a third frequency that is the same as the second frequency.

11. The device of claim **1**, wherein the processor is configured to receive the second data in the second time slot on a third frequency that is the same as the first frequency.

12. A wireless battery management system (WBMS), comprising:

a first primary device configured to wirelessly broadcast first data during a first time slot; and

a secondary device configured to:

wirelessly receive the first data during the first time slot;

wirelessly receive second data broadcasted by a second primary device during a second time slot after the first time slot;

wirelessly transmit third data to the first primary device during a third time slot after the second time slot, the third data comprising a battery cell status; and

wirelessly transmit the third data to the second primary device during a fourth time slot after the second time slot.

13. The system of claim **12**, wherein the first and second time slots are consecutive time slots.

14. The system of claim **12**, wherein the first and second data together include timing information for the third and fourth time slots.

15. The system of claim **12**, wherein the secondary device is configured to store timing information for the third and fourth time slots prior to receiving the first data and prior to receiving the second data.

16. The system of claim **12**, wherein the secondary device is configured to transmit the third data in the third time slot on a first frequency and the third data in the fourth time slot on a second frequency different from the first frequency using a frequency hopping scheme.

17. The system of claim **12**, wherein the secondary device is a first secondary device, the system further comprising a second secondary device configured to transmit fourth data to the first primary device during the fourth time slot.

18. The system of claim **17**, wherein the second secondary device is configured to transmit the fourth data to the second primary device during the third time slot.

19. A non-transitory computer readable medium storing instructions which, when executed by a processor, cause the processor to:

during a first time slot, receive first data wirelessly transmitted by a first primary device to the processor and to a secondary device;

during a second time slot, cause wireless transmission of second data to the first primary device simultaneous with transmission of third data from the secondary device to a second primary device, the second and third data comprising battery cell status; and

during a third time slot, cause wireless transmission of the second data to the second primary device simultaneous with transmission of the third data from the secondary device to the first primary device.

20. The medium of claim **19**, wherein the instructions cause the processor to transmit the second data to the first primary device and to the second primary device on different frequencies.

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